Computer Systems Modelling: Lent 2023 **Assessment 2**
Instructor: Prof. S. Keshav 10-11 AM, March 16, 2023
Answer all questions Maximum marks 15

Question 1. Use Euler's identity to prove that

$$sin(A+B) = sinAcosB + cosAsinB$$

(1 mark)

Solution: The right hand side is

$$\frac{(e^{jA} - e^{-jA})(e^{jB} + e^{jB})}{(2)(2j)} + \frac{(e^{jA} + e^{-jA})(e^{jB} - e^{jB})}{(2)(2j)}$$

which reduces trivially to

$$\frac{2e^{j(A+B)} - 2e^{-j(A+B)}}{4j} = \frac{e^{j(A+B)} - e^{-j(A+B)}}{2j} = sin(A+B)$$

Question 2. Compute the convolution of a function that has the constant value of 1 in the interval [-1, 1] with the function x^2 clipped to interval [0,1].**(3 marks)**

Solution: Let $a(t)$ denote the clipped quadratic and let $b(t)$ denote the constant function. We want to compute $a(t) \otimes b(t)$, which is defined as

$$a(t) \otimes b(t) = \int_0^\infty a(\tau)b(t-\tau)d\tau \tag{1}$$

Note that $b(\tau)$ is symmetric about the Y axis, so this can be rewritten as:

$$a(t) \otimes b(t) = \int_0^\infty a(\tau)b(\tau-t)d\tau \tag{2}$$

which we recognize as time-shifting $b(t)$ by t. Thus, the integral can be reduced to

$$\begin{cases} \int_0^{t+1} \tau^2 d\tau \text{ for } -1 \leq t < 0 \\ \int_0^1 \tau^2 d\tau \text{ for } 0 \leq t < 1 \\ \int_{t-1}^1 \tau^2 d\tau \text{ for } 1 \leq t \leq 2 \end{cases}$$

$$\begin{cases} \frac{\tau^3}{3}\Big|_0^{t+1} \text{ for } -1 \leq t < 0 \\ \frac{\tau^3}{3}\Big|_0^1 \text{ for } 0 \leq t < 1 \\ \frac{\tau^3}{3}\Big|_{t-1}^1 \text{ for } 1 \leq t \leq 2 \end{cases}$$

$$\begin{cases} \frac{(t+1)^3}{3} \text{ for } -1 \leq t < 0 \\ \frac{1}{3} \text{ for } 0 \leq t < 1 \\ \frac{(t-1)^3}{3} \text{ for } 1 \leq t \leq 2 \end{cases}$$

Question 3. What is the kth term of the Fourier series corresponding to the eternal signal $(cost + sin2t)$ **(2 marks)**

Solution: Since the first term is a harmonic of the second, the period is 2π and $\omega_0 = 2\pi/T_0 = 1$.

$$c_k = \frac{1}{2\pi}\int_0^{2\pi}(cost + sin2t)e^{-jkt}dt$$

$$= \frac{1}{2\pi.2}\int_0^{2\pi}(e^{jt} + e^{-jt})e^{-jkt}dt + \frac{1}{2\pi.2j}\int_0^{2\pi}(e^{j2t} - e^{-j2t})e^{-jkt}dt$$

$$= \frac{1}{4\pi}\left[\int_0^{2\pi} e^{j(1-k)t}dt + \int_0^{2\pi} e^{-j(1+k)t}dt\right] + \frac{1}{4\pi j}\left[\int_0^{2\pi} e^{j(2-k)t}dt + \int_0^{2\pi} e^{-j(2+k)t}dt\right]$$

$$= \frac{1}{4\pi}\left[\frac{e^{j(1-k)t}}{j(1-k)}\Big|_0^{2\pi} + \frac{e^{-j(1+k)t}}{-j(1+k)}\Big|_0^{2\pi}\right] + \frac{1}{4\pi j}\left[\frac{e^{j(1-k)t}}{j(2-k)}\Big|_0^{2\pi} + \frac{e^{-j(1+k)t}}{-j(2+k)}\Big|_0^{2\pi}\right]$$

$$= \frac{1}{4\pi}\left[\frac{e^{j(1-k).2\pi}}{j(1-k)} + \frac{e^{-j(1+k).2\pi}}{-j(1+k)}\right] + \frac{1}{4\pi j}\left[\frac{e^{j(1-k).2\pi}}{j(2-k)} + \frac{e^{-j(1+k).2\pi}}{-j(2+k)}\right]$$

Question 4. Use the integral rule to find the Laplace transform of the function t^2. **(1 mark)**

Solution: This is $\int\int u(t)dt = L(u(t))/s^2 = 1/s^3$.

Question 5. Suppose that the throughput of a network flow, T, is related to the TCP buffer size at an endpoint, the control variable B, by the equation:

$$2\frac{dT^2}{dB} + 3\frac{dT}{dB} + T = B$$

What is the transfer function? (1 mark) What is the time evolution the system (1 mark) Is this system stable? Why or why not? (1 mark) **(Total 3 marks)**

Solution: Taking the Laplace transform of both sides

$$2s^2T(s) + 3sT(s) + T(s) = B(s)$$

$$T(s) = \frac{B(s)}{2s^2 + 3s + 1}$$

So, the transfer function is

$$\frac{1}{2s^2 + 3s + 1}$$

$$= \frac{1}{(2s+1)(s+1)} = \frac{2}{2s+1} - \frac{1}{2(s+1)}$$

This immediately tells us that the time evolution of the system is of the form

$$c_1e^{-1/2t} - c_2e^{-t}$$

from the linearity of the Laplace transform and the fact that $\frac{K}{s+a} \Longleftrightarrow e^{-at}$.
Both roots of the transfer function (-1 and -1/2) are in the left half plane, so the system is stable.

Question 6. In an underdamped system, we find that the response to a unit step input at time 0 is a decaying sinusoid with a phase shift of $\pi/3$ and a natural frequency of 2π rad/s, asymptoting to a unit step. What is the transfer function? **(2 marks)**

Solution: We have $\omega_d = 2\pi$, $\zeta = cos^{-1}\pi/3 = .5$ and $K = 1$. ω_d and ω_n are related by $\omega_d^2 = \omega_n^2(1 - \zeta^2)$, so

$$\omega_n = \frac{4\pi}{\sqrt{3}}$$

the transfer function is

$$\frac{1}{(\frac{3s^2}{16\pi^2}) + \frac{\sqrt{3}s}{4\pi} + 1}$$

Question 7.

a. What is the output $y(t)$ of a controlled plant with transfer function $G(s) = \frac{1}{s+a}$ when the controller is defined by $U(s) = bs(R–Y)$, the command R is the unit step function, and $W = 0$. (2 marks)

b. What is the steady state error given a step input to this controller? (1 mark)

(Total 3 marks)

Solution: $Y = G(s)U(s) = \frac{bs(R-Y)}{s+a}$, so $Y = \frac{bs}{(1+b)s+a}R$. For a step input $R(s) = \frac{1}{s}$, and $Y = \frac{b}{(1+b)s+a}$ $= (\frac{b}{1+b})(\frac{1}{s+\frac{a}{1+b}})$, which is of the form $\frac{K}{s+a}$, so the inverse transform is

$$y(t) = \frac{b}{1+b}e^{-\frac{at}{1+b}}$$

For a step input where $R(s) = 1/s$, we have already shown that $Y = \frac{b}{(1+b)s+a}$. The final value theorem states that the steady state error is given by

$$\lim_{s\to 0} Y(s) = \lim_{s\to 0} \frac{b}{(1+b)s+a} = b/a$$